Floris P. Barthel, M.D., Ph.D.

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PROFESSIONAL APPOINTMENTS

- The University of Arizona (UA) • *Adjunct Professor* • 2021-present
  Department of Basic Medical Sciences at the College of Medicine
- Arizona State University (ASU) • *Adjunct Professor* • 2021-present
  School of Life Sciences and College of Liberal Arts and Sciences
- the Molecular and Cellular Biology of Cancer program • *Program Member* • 2021-present
  City of Hope's NCI-designated Comprehensive Cancer Center
- the Beckman Research Institute at City of Hope • *Assistant Professor* • 2021-present
  Department of Computational and Quantitative Medicine
- the Translational Genomics Research Institute (TGen) • *Assistant Professor* • 2021-present
  Bioinnovation and Genome Sciences

TRAINING AND EDUCATION

- Jackson Laboratory for Genomic Medicine • Postdoctoral Associate • 2016-2021
- Vrije Universiteit Amsterdam • Ph.D (cum laude) • 2014-2022
- Vrije Universiteit Amsterdam Medical Center • Medicine (M.D.) • 2011-2014
- Vrije Universiteit Amsterdam Medical Center • Medicine (B.Sc.) • 2007-2011

PREPRINTS

** denotes first authorship, † denotes corresponding authorship*

- Julian K. Lucas, …, **Floris P. Barthel**, …, Benedict Paten. *HPRC2: A human pangenome reference with near-complete coverage of common genetic variation*. **bioRxiv**. 2026 Jul 22. doi: 10.64898/2026.07.21.739710.
- Ogechukwu Mbegbu, …, **Floris P Barthel**†. *Telomere Dysfunction in Human Astrocytes Drives Acrocentric Chromosome Instability and Nucleolar Reorganization*. **bioRxiv**. 2026 May 21. doi: 10.64898/2026.05.19.726354.
- T. Rhyker Ranallo-Benavidez, …, **Floris P. Barthel**†. *KaryoScope: rapid, alignment-free sequence annotation for the pangenome era*. **bioRxiv**. 2026 May 17. doi: 10.64898/2026.05.15.725544.
- Justin Wagner*, Ayse G. Keskus*, Keisuke K. Oshima*, T. Rhyker Ranallo-Benavidez*, …, **Floris P. Barthel**†, Chunlin Xiao†, Glennis A. Logsdon†, Mikhail Kolmogorov†, Justin M. Zook†. *A complete human pancreatic cancer genome*. **bioRxiv**. 2026 May 6. doi: 10.64898/2026.05.01.722316.
- Danilo Dubocanin, …, **Floris P. Barthel**, …, Andrew B. Stergachis. *Single-molecule variation in telomeric sequence and structure across humans*. **bioRxiv**. 2026 May 5. doi: 10.64898/2026.05.01.722200.
- Maryam Jehangir, …, **Floris P. Barthel**†. *Chromosome-arm-specific telomere length governs dual modes of structural genome evolution in IDH-mutant astrocytoma*. **bioRxiv**. 2026 Apr 23. doi: 10.64898/2026.04.22.720226.
- Jarno N. Alanko, …, **Floris P. Barthel**, …, Camille Marchet. *Hierarchical genomic feature annotation with variable-length queries*. **bioRxiv**. 2026 Mar 18. doi: 10.64898/2026.03.15.711907.
- Yi-An Chen, …, **Floris P Barthel**†. *Mapping the Telomeric 3D Interactome with Telomere-C Reveals Repetitive Element Hubs Associated with Telomere Maintenance*. **bioRxiv**. 2025 Oct 10. doi: 10.1101/2025.10.09.681516.
- Sharvari Mankame, …, **Floris Barthel**†. *Cell-Free DNA Reveals the Longitudinal Effects of Temozolomide Treatment and Host Co-Culture in Glioblastoma Models*. **bioRxiv**. 2025 Jul 12. doi: 10.1101/2025.07.12.662613.
- Sunada Khadka, …, **Floris P Barthel**, …, Florian L Muller. *Anaplerotic nutrient stress drives synergy of angiogenesis inhibitors with therapeutics targeting tumor metabolism*. **bioRxiv**. 2023 May 9. doi: 10.1101/2023.05.07.539744.

PEER REVIEWED PUBLICATIONS

** denotes first authorship, † denotes corresponding authorship*

- T Rhyker Ranallo-Benavidez, . . . , **Floris P Barthel**. *Haplotype-resolved genome assemblies of BJ and IMR-90 human fibroblast cell lines reveal extensive structural variation and enable reanalysis of historical sequencing data*. **Nucleic Acids Research**. 2026 Apr 23;54(8):gkag333. doi: 10.1093/nar/gkag333. PubMed PMID: 42049241. PubMed Central PMCID: PMC13124242.
- Jimin Park, . . . , **Floris P Barthel**, . . . , Kishwar Shafin. *Accurate somatic small variant discovery for multiple sequencing technologies with DeepSomatic*. **Nature Biotechnology**. 2025 Oct 16;1-10. doi: 10.1038/s41587-025-02839-x. PubMed PMID: 41102444.
- **Floris P Barthel***†. *Uncommon territory: The clinical and molecular profile of metastatic gliomas*. **Neuro-oncology**. 2025 Oct 15;noaf244. doi: 10.1093/neuonc/noaf244. PubMed PMID: 41092320.
- Tathiane M Malta, . . . , Houtan Noushmehr. *The Epigenetic Evolution of Glioma Is Determined by the IDH1 Mutation Status and Treatment Regimen*. **Cancer Research**. 2024 Mar 4;84(5):741-756. doi: 10.1158/0008-5472.CAN-23-2093. PubMed PMID: 38117484. PubMed Central PMCID: PMC10911804.
- Aram S Modrek, . . . , **Floris P Barthel**, . . . , Erik P Sulman. *PDPN marks a subset of aggressive and radiation-resistant glioblastoma cells*. **Frontiers in Oncology**. 2022 Aug 10;12:941657. doi: 10.3389/fonc.2022.941657. PubMed PMID: 36059614. PubMed Central PMCID: PMC9434399.
- Frederick S Varn, . . . , **Floris P Barthel**, . . . , et al. (GLASS Consortium). *Glioma progression is shaped by genetic evolution and microenvironment interactions*. **Cell**. 2022 May 31;185(12):2184-2199.e16. doi: 10.1016/j.cell.2022.04.038. PubMed PMID: 35649412. PubMed Central PMCID: PMC9189056.
- Kevin C Johnson, . . . , **Floris P Barthel**, . . . , Roel GW Verhaak. *Single-cell multimodal glioma analyses identify epigenetic regulators of cellular plasticity and environmental stress response*. **Nature Genetics**. 2021 Sep 30;53(10):1456-1468. doi: 10.1038/s41588-021-00926-8. PubMed PMID: 34594038. PubMed Central PMCID: PMC8570135.
- Niels Verburg*, **Floris P Barthel***, . . . , Roel GW Verhaak. *Spatial concordance of DNA methylation classification in diffuse glioma*. **Neuro-oncology**. 2021 May 28;23(12):2054-2065. doi: 10.1093/neuonc/noab134. PubMed PMID: 34049406. PubMed Central PMCID: PMC8643482.
- Emre Kocakavuk, . . . , **Floris P Barthel**†, Roel GW Verhaak. *Radiotherapy is associated with a deletion signature that contributes to poor outcomes in patients with cancer*. **Nature Genetics**. 2021 May 27;53(7):1088-1096. doi: 10.1038/s41588-021-00874-3. PubMed PMID: 34045764. PubMed Central PMCID: PMC8483261.
- Patrick Y Wen, . . . , **Floris P Barthel**, . . . , Martin J Van den Bent. *Glioblastoma in adults: a Society for Neuro-Oncology (SNO) and European Society of Neuro-Oncology (EANO) consensus review on current management and future directions*. **Neuro-oncology**. 2020 May 19;22(8):1073-1113. doi: 10.1093/neuonc/noaa106. PubMed PMID: 32328653. PubMed Central PMCID: PMC7594557.
- Samirkumar B Amin, . . . , **Floris P Barthel**, . . . , Roel GW Verhaak. *Comparative molecular life history of spontaneous canine and human gliomas*. **Cancer Cell**. 2020 Feb 10;37(2):243-257.e7. doi: 10.1016/j.ccell.2020.01.004. PubMed PMID: 32049048. PubMed Central PMCID: PMC7132629.
- Kevin J Anderson, . . . , **Floris P Barthel**, . . . , Mustafa Khasraw. *Molecular and clonal evolution in recurrent metastatic gliosarcoma*. **Molecular Case Studies**. 2020 Feb 3;6(1):a004671. doi: 10.1101/mcs.a004671. PubMed PMID: 31896544. PubMed Central PMCID: PMC6996521.
- **Floris P Barthel***, . . . , Roel GW Verhaak. *Longitudinal molecular trajectories of diffuse glioma in adults*. **Nature**. 2019 Nov 20;576(7785):112-120. doi: 10.1038/s41586-019-1775-1. PubMed PMID: 31748746. PubMed Central PMCID: PMC6897368.
- **Floris P Barthel***, . . . , Roel GW Verhaak. *Evolving insights into the molecular neuropathology of diffuse gliomas in adults*. **Neurologic Clinics**. 2018 Aug;36(3):421. doi: 10.1016/j.ncl.2018.04.002. PubMed PMID: 30072063. PubMed Central PMCID: PMC6322909.
- Francesca Menghi, **Floris P Barthel**, . . . , Edison T Liu. *The tandem duplicator phenotype is a prevalent genome-wide cancer configuration driven by distinct gene mutations*. **Cancer Cell**. 2018 Jul 12;34(2):197-210.e5. doi: 10.1016/j.ccell.2018.06.008. PubMed PMID: 30017478. PubMed Central PMCID: PMC6481635.
- **Floris P Barthel***†, Pieter Wesseling, Roel GW Verhaak. *Reconstructing the molecular life history of gliomas*.

Acta Neuropathologica. 2018 May 3;135(5):649-670. doi: 10.1007/s00401-018-1842-y. PubMed PMID: 29616301. PubMed Central PMCID: PMC5904231.

- Xin Hu, . . . , **Floris Barthel**, . . . , Roel GW Verhaak. *TumorFusions: an integrative resource for cancer-associated transcript fusions*. **Nucleic Acids Research**. 2018 Jan 4;46(D1):D1144-D1149. doi: 10.1093/nar/gkx1018. PubMed PMID: 29099951. PubMed Central PMCID: PMC5753333.
- Alexander J Lazar, . . . , incl. **Floris P Barthel**, . . . , Xin Hu). *Comprehensive and integrated genomic characterization of adult soft tissue sarcomas*. **Cell**. 2017 Nov 2;171(4):950-965. doi: 10.1016/j.cell.2017.10.014. PubMed PMID: 29100075. PubMed Central PMCID: PMC5693358.
- Qianghu Wang, . . . , **Floris Barthel**, . . . , Roel GW Verhaak. *Tumor evolution of glioma-intrinsic gene expression subtypes associates with immunological changes in the microenvironment*. **Cancer Cell**. 2017 Jul 10;32(1):42-56.e6. doi: 10.1016/j.ccell.2017.06.003. PubMed PMID: 28697342. PubMed Central PMCID: PMC5599156.
- Xin Hu, . . . , **Floris Barthel**, . . . , Roel GW Verhaak. *Multigene signature for predicting prognosis of patients with 1p19q co-deletion diffuse glioma*. **Neuro-oncology**. 2017 Jun;19(6):786-795. doi: 10.1093/neuonc/now285. PubMed PMID: 28340142. PubMed Central PMCID: PMC5464432.
- **Floris P Barthel***, . . . , Roel GW Verhaak. *Systematic analysis of telomere length and somatic alterations in 31 cancer types*. **Nature Genetics**. 2017 Jan 30;49(3):349-357. doi: 10.1038/ng.3781. PubMed PMID: 28135248. PubMed Central PMCID: PMC5571729.
- Michele Ceccarelli*, **Floris P Barthel***, Tathiane M Malta*, Thais S Sabedot*, . . . , et al. (TCGA Research Network). *Molecular profiling reveals biologically discrete subsets and pathways of progression in diffuse glioma*. **Cell**. 2016 Jan 28;164(3):550-563. doi: 10.1016/j.cell.2015.12.028. PubMed PMID: 26824661. PubMed Central PMCID: PMC4754110.

HONORS AND AWARDS

- ASCI Young Physician-Scientist Award • 2022
- Sylvia-Chase Foundation Award • 2021
- Brain Tumor Charity Celebrating You Research Rising Star Award • 2020
- William Guy Forbeck Research Foundation Scholar Award • 2019
- Dutch Cancer Society (KWF) Research Intern Award • 2014
- Foundation of Renswoude Research Intern Award • 2014
- European Society for Neurological Surgery Travel Award • 2013

RESEARCH SUPPORT

- 1R21NS142788 (NIH) • Role: PI • 04/15/2026–03/31/2028
  Molecular cartography of initiating and infiltrating glioma cells using multisector sampling and spatial transcriptomics
- 2025-1600 (FAP Research Initiative) • Role: Co-I • 12/01/2025–11/30/2028
  Beyond APC: Mapping Genetic and Epigenetic Determinants of Familial Adenomatous Polyposis
- W81XWH2120009 (DOD) • Role: PI • 09/15/2025–09/14/2026
  Identifying Uncharted Cytogenetic Features of Aggressive Thyroid Cancer using Computational Karyotypes
- CLIN2-13162 (CIRM) • Role: Co-I • 07/15/2025–11/30/2026
  A Phase I Study of Multiple doses of NSC-Based Oncolytic Virotherapy Administered Intracerebrally to Patients with Recurrent High-Grade Gliomas
- RFGA-2024-022-011 (ABRC) • Role: PI • 03/01/2025–02/29/2028
  Tracking tumor burden and response to IL13Rα2 CAR-T therapy in glioblastoma patients using CSF-based liquid biopsies
- R01NS134116 (NIH) • Role: Co-I • 09/01/2024–08/31/2027
  Improving Glioma Immunotherapy Efficacy by Regulating Tumor Inflammation
- HT9425-24-1-0195 (DOD) • Role: Co-I • 04/01/2024–03/31/2026

The role of TDP-43 associated cryptic exon inclusion in KALRN on C9orf72-mediated cortical neurodegeneration

- Lennar Foundation • Role: PI • 09/01/2023–08/31/2026
  Addressing a Persistent Barrier in Cancer Care: Detecting and Treating Aggressive Cancers
- HT9425-23-1-0844 (DOD) • Role: PI • 08/01/2023–07/31/2027
  Genomic Dissection of Structural Mutations Following Telomere Dysfunction in Glioblastoma Cells and Patients
- DG2200042 (ABTA) • Role: PI • 10/01/2022–09/30/2023
  Accurate Glioma Burden Tracking Using Circulating Tumor Mitochondrial DNA
- R00CA226387 (NIH) • Role: PI • 08/01/2021–07/31/2024
  Genetic and Epigenetic Strategies for the Acquisition of Telomere Maintenance in Human Cancer Cells

INVITED TALKS

- *Alternative ends to human ALT cancer chromosomes*. Invited talk at the François Jacob Institute of Biology, CEA; 2026 Sep 21; Fontenay-aux-Roses, France.
- *Genomics has no one sun*. Invited talk at the International Workshop on Chromosomal Instability; 2026 Sep 16-18; Segovia, Spain.
- *Unusual chromosome ends in IDH mutant astrocytoma*. Invited talk at Cancer Center Amsterdam; 2026 Jul 31; Amsterdam, The Netherlands.
- *Break-induced replication repositions ribosomal DNA arrays to chromosome ends in response to telomere crisis*. Invited talk at the Netherlands Cancer Institute; 2026 Jun 10; Amsterdam, The Netherlands.
- *Break-induced replication repositions ribosomal DNA arrays to chromosome ends in response to telomere crisis*. Invited talk at the Bioinformatics & Genome Biology Lecture Series; 2026 Jun 5; Rome, Italy.
- *Beyond the breakage-fusion-bridge cycle: telomere dysfunction drives long-range satellite DNA recombination events through spatial proximity*. Invited talk at the Seminar Series; 2026 May 4; New Haven, CT.
- *In-silico cytogenetic analysis of the Human Pangenome reveals unexpected D4Z4 array diversity and centromeric satellite polymorphisms*. Virtual invited talk at HPRC-all consortium invited presentation; 2026 Feb 11; Zoom.
- *Beyond the breakage-fusion-bridge cycle: telomere dysfunction drives long-range satellite DNA recombination events through spatial proximity*. Invited talk at the Cancer Data Science Laboratory Seminar; 2026 Jan 19; Bethesda, MD.
- *Evaluating immunotherapy response and resistance in real-time using cell-free DNA*. Invited talk at the AARN meeting; 2025 Sep 14; Schaumburg, IL.
- *Acrocentric chromosome arm fusions in transformed astrocytes*. Invited talk at the RSO Advance Annual Conference; 2025 Apr 11; Lake Arrowhead, CA.
- *Resolving structure and function of haplotype-phased sex chromosomes in glioblastoma*. Invited talk at the Sex Differences P01 consortium meeting; 2024 Sep 25; St Louis, MO.
- *Resolving the genotype and phenotype of a "simple" engineered structural variant*. Invited talk at the Forbeck Forum on The Mechanisms of Cancer Genome Rearrangements; 2024 Aug 20; Aspen, CO.
- *Genomic technology development for translational applications in the management of adult diffuse glioma*. Invited talk at the Banner Health Neurosurgery seminar series; 2024 Apr 11; Phoenix, AZ.
- *Disentangling long-range telomeric chromatin interactions: implications in cancer and aging*. Virtual invited talk at the City of Hope Cancer Center Joint CGE-RadOnc Research Forum; 2022 May 11; Zoom.
- *The Molecular Life History of Adult Gliomas*. Virtual invited talk at the Barrow Neurological Institute Neuroscience Conference; 2022 Jan 18; Zoom.
- *Reconstructing the Molecular Life History of Gliomas*. Invited talk at the AZ Regenerative Medicine Symposium; 2021 Nov 18; Tempe, AZ.
- *Clinical Impact of Molecular Neuro-Oncology*. Virtual invited talk at the Maine Cancer Genomics Initiative Forum; 2021 Mar 26; Webex.

CONFERENCE PRESENTATIONS

- **Barthel FP**. *Break-induced replication repositions ribosomal DNA and alpha-satellite arrays to chromosome ends in human ALT cancers*. Oral presentation at the Federation of American Societies for Experimental Biology (FASEB 2026); 2026 Sep 28; Melbourne, Florida.
- **Barthel FP**. *When T2T is incomplete: ribosomal DNA and alpha-satellite arrays cap the ends of a subset of human ALT chromosomes*. Oral presentation at T2T-F2F "Face-to-Face" (2026); 2026 Sep 3; Santa Cruz, California.
- **Barthel FP**. *When T2T is incomplete: functional ribosomal DNA arrays mark the ends of a subset of human ALT chromosomes*. Oral presentation at IGGSy (2026); 2026 Jul 6.
- **Barthel FP**. *Functional ribosomal DNA arrays mark the ends of a subset of human ALT chromosomes*. Poster presentation at EMBO Telomeres (2026); 2026 May 28.
- **Barthel FP**. *Break-induced replication drives telomeres to recombine with DNA satellites during telomere crisis*. Oral presentation at EMBO Telomeres (2026); 2026 May 28.
- **Barthel FP**. *Break-induced replication drives telomeres to recombine with DNA satellites during telomere crisis*. Oral presentation at Biology of Genomes (2026); 2026 May 5.
- **Barthel FP**. *Telomere dysfunction promotes remote recombination events between telomeres, centromeres and ribosomal DNA that are catalyzed by long-range chromatin interactions*. Oral presentation at the American Society of Human Genetics (2025); 2025 Oct 18; Boston, Massachusetts.
- **Barthel FP**. *Resolving the genotype and phenotype of a "simple" engineered structural variant*. Oral presentation at Mechanisms of Cancer Genome Rearrangements; 2024 Sep 21; Aspen, Colorado.
- **Barthel FP**. *Telomeres and 3D Genome Organization*. Poster presentation at Chromatin dynamics and nuclear organization in genome maintenance; 2024 Jun 18; Leiden, Netherlands.
- **Barthel FP**. *Stereotactic image-guided epigenome profiling reveals diverse evolutionary growth routes in diffuse gliomas*. Oral presentation at the Society for Neuro-Oncology (2020); 2020 Nov 19.
- **Barthel FP**. *Longitudinal Molecular Trajectories of Diffuse Glioma in Adults*. Oral presentation at the AACR Special Conference on Evolutionary Dynamics in Carcinogenesis and Response to Therapy (2020); 2020 Mar 12.
- **Barthel FP**. *Longitudinal Molecular Trajectories of Diffuse Glioma in Adults*. Oral presentation at the Keystone Symposium on Cancer Evolution and Combinatorial Cancer Therapies (2020); 2020 Jan 19.
- **Barthel FP**. *Longitudinal Molecular Trajectories of Diffuse Glioma in Adults*. Oral presentation at the Society for Neuro-Oncology (2019); 2019 Nov 20.
- **Barthel FP**. *Next-generation assay development for probing telomeres in cancer*. Oral presentation at the Forbeck Forum on Telomerase-Mediated Telomere Targeting in Cancer (2019); 2019 Oct 17.
- **Barthel FP**. *A genomic basis for cancer immortalization: a computational biologist's perspective*. Oral presentation at the MD Anderson Symposia on Cancer Research, Cancer Evolution: Mechanisms of Vulnerability and Resistance (2016); 2016 Oct 6.
- **Barthel FP**. *A pan-glioma characterization of genomic epigenomic and transcriptomic activities reveals novel relationships between histological subtypes and molecular signature*. Oral presentation at the Society for Neuro-Oncology (2015); 2015 Nov 19.

OVERVIEW OF CURRENT AND PAST TRAINEES

- Divyansh • Feb 2026 - present • Graduate Student
- Griffin Azrak • Feb 2026 - present • Graduate Student
- Daniel Chambers • Aug 2025 - present • Graduate Student
- Nicholas Costigan • Feb 2025 - present • Graduate Student
- Shea Gallus • Oct 2024 - present • Clinical Fellow
- Maryam Jehangir • Apr 2024 - present • Postdoc
- Yue Hao • Jan 2024 - present • Computational Scientist

- T. Rhyker Ranallo-Benavidez • May 2023 - present • Computational Scientist II
- Sharvari Mankame • Aug 2022 - present • Graduate Student
- Mimi Mbegbu • May 2022 - present • Research Associate III
- Noelle Fukushima • Sep 2021 - present • Senior RA
- Yi-An Chen • Sep 2021 - present • Computational Scientist
- Charles Cickavage • Oct 2024 • Intern
- Rebecca Reiman • Oct 2024 • Research Staff
- Michelle Padilla Soto • Jun 2024 • Helios Scholar
- Ryan O'Hanrahan • Dec 2023 • Undergrad
- Manasa Iyer • Jun 2023 • Helios Scholar
- Hersh Nanda • Jun 2023 • Helios Scholar
- Maria Kyriakidou • Nov 2022 • Postdoc
- Victoria Jimenez • Jun 2022 • Undergrad
- Sarah Weiss • Jun 2022 • Undergrad
- Shuvodeep De • Dec 2021 • Postdoc
- Marisol Araiza • Dec 2021 • Undergrad
- Abubakr El Sobky • Nov 2021 • Undergrad
- Riya Naik • May 2021 • Undergrad

JOURNAL PEER REVIEW

- Acta Neuropathologica • 1 review
- Bioinformatics • 2 reviews
- Cancer Cell • 3 reviews
- Cancer Discovery • 1 review
- Cancers • 2 reviews
- Cell Reports • 1 review
- Clinical Cancer Research • 2 reviews
- Communications Biology • 1 review
- Current Opinion in Genetics & Development • 1 review
- eLife • 2 reviews
- International Journal of Cancer • 1 review
- JCO Precision Oncology • 1 review
- Journal of Translational Medicine • 1 review
- Molecular Cell • 4 reviews
- Nature Communications • 2 reviews
- Neuro-Oncology • 12 reviews
- Science Advances • 3 reviews
- Scientific Reports • 1 review

AD HOC GRANT REVIEW

- Congressionally Directed Medical Research Programs (CDMPR), Rare Cancers Research Program • 2025

- KU Leuven, C2 award • 2024
- Brain Research UK, • 2024
- American Brain Tumor Association, • 2024
- Congressionally Directed Medical Research Programs (CDMPR), Glioblastoma Research Program • 2024
- NIH/NIEHS, Special Emphasis Panel ZES1 LWJ K (TL) • 2023
- Brain Tumour Charity, • 2022
- Brain Research UK, • 2022

PROFESSIONAL MEMBERSHIPS

- American Society of Human Genetics, Member • 2019-present
- American Association for Cancer Research, Active Member • 2017-present
- Society for Neuro Oncology, Member • 2015-present